Optimal strategies in the all-heads coin game

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Abstract

We study a sequential coin-flipping game in which a player starts with n coins, each landing heads independently with probability p. In each round the player flips all remaining coins and must set aside at least one coin showing heads; if no coin shows heads, the player loses. The player wins if and when all coins have been set aside. This is a stochastic shortest-path Markov decision process whose Bellman equation involves a nonlinear suffix-maximum operator. We analyse two natural strategies — One (set aside exactly one head) and All (set aside every head) — and determine the optimal winning probability $w_{n,p}$ as a function of n and p.

For $p = \frac{1}{2}$ every strategy achieves $w_{n,1/2} = \frac{1}{2}$. For $p > \frac{1}{2}$ the strategy One is optimal and $n \mapsto w_{n,p}$ is strictly increasing; as a consequence the limit $W(p) := \lim_n w_{n,p}$ admits an explicit series representation and satisfies $p \le W(p) < 1$. For $p < \frac{1}{2}$ near $\frac{1}{2}$ we carry out a first-order perturbation expansion in $\delta := \frac{1}{2} - p$ and prove that the deficit is $\frac{1}{2} - w_{n,\frac{1}{2}-\delta} \approx \delta c_n$ in first order in δ, where c_n satisfies a linear recursion for $n \ge 7$, with a limit $L = \lim c_n \approx 1.7035$. As a consequence the optimal value sequence has, to first order in δ, a strict local minimum at $n = 5$ and no local maximum — local maxima are a non-perturbative phenomenon.

1 Introduction

A note on authorship. The mathematical text, the numerical code (`simulation/`), and the complete Lean 4/Mathlib formalization in this paper were produced by Anthropic's Claude (versions Opus 4.6 and Opus 4.7), interactively, in conversations with the author. To be precise about attribution: the underlying mathematical ideas, the selection of the research question (alternative objective $w_{n,p}$ vs. $v_{n,p}$), the decision to carry out a first-order perturbation in $\delta = \frac{1}{2} - p$, the structuring of §4 as a joint induction, and the decision to formally verify the paper are all the author's. Claude's role was execution: drafting the exposition from the author's outline, proposing and debugging proof tactics in Lean, selecting Mathlib lemmas, spotting where a generic approach had to be replaced by a case-by-case one (for instance the cubic-bound inequality in §4.3, or the Tannery-style dominated convergence argument in the proof of Theorem 4.10), and producing the numerical scripts. Individual decisions in both categories were settled through back-and-forth dialogue; the author reviewed every edit.

Readers who want the full record can consult a prompt-by-prompt transcript at `journal.md`; a more granular development history of the formalization appears in Appendix A.

1.1 The game

Fix a probability $p \in (0,1)$ and a positive integer n. A player is given n coins, each of which independently lands heads with probability p and tails with probability $q := 1-p$. The game proceeds in rounds. In each round:

(i) All remaining coins are flipped simultaneously.

(ii) If no coin shows heads, the game is lost.

(iii) Otherwise the player must set aside at least one coin that shows heads. The set-aside coins leave the game permanently.

(iv) If every coin has been set aside, the game is won. Otherwise, a new round begins with the remaining coins.

The player's objective is to maximise the probability of winning.

Two natural strategies present themselves:

- **Strategy One:** set aside exactly one head when $1 \le k \le n-1$ heads appear, and set aside all n when $k = n$.
- **Strategy All:** set aside every head that appears ($i = k$).

Strategy One is conservative: it preserves as many coins as possible for future rounds, betting that the one set-aside coin has already "done its job." Strategy All is greedy: it cashes in every success immediately. Neither is universally optimal.

Relation to prior work. A close variant of this game was introduced and studied by van Doorn [7], who took as objective the *expected number of heads* collected, $v_{n,p}$, rather than the all-heads-wins probability $w_{n,p}$ studied here. Van Doorn's analysis is essentially complete for $p \ge \frac{1}{2}$: he proves that there exist absolute constants $\phi = (\sqrt{5}-1)/2 \approx 0.618$ and $p_0 \approx 0.5495$ such that the optimal strategy follows One (in his notation, "A") for $p \ge \phi$, switches between One and All at a level $n(p)$ for $p_0 < p < \phi$, and follows All ("B") throughout for $\frac{1}{2} \le p \le p_0$. For $p < \frac{1}{2}$, however, van Doorn writes that "everything is a lot less clear" and restricts himself to numerical conjectures and a non-monotonicity result.

The present paper makes two changes in emphasis. First, the objective: maximising the all-heads probability $w_{n,p}$ rather than the expected count $v_{n,p}$. As a consequence, the $p > \frac{1}{2}$ case becomes simpler: the strategy One is always optimal — a fact noted in passing in [7, §10] and proved here in Theorem 3.1 below with a self-contained joint induction. Second, and more importantly, our focus is the regime $p < \frac{1}{2}$ that van Doorn explicitly leaves open. We carry out a first-order perturbation in $\delta = \frac{1}{2} - p$ and obtain a *closed-form description* of the leading behaviour: the deficit coefficient c_n satisfies a linear recursion (Proposition 4.8), the limit $L = \lim c_n$ admits an explicit formula and a numerical value $L \approx 1.7035$ (Theorem 4.10), and the local-extremum structure of $n \mapsto w_{n,p}$ is fully determined to first order (Corollary 4.11).

A third, methodological, change concerns formal verification. Since the results of the present paper were generated in collaboration with AI, we want to avoid hallucinations: the reader does not have to be a specialist in Lean 4, the corresponding proof assistant, but

has to acknowledge the fact that (i) the recursive definition for $w_{n,p}$ from (1) is correctly implemented and (ii) the results as they stand compile without error, meaning that Lean 4 has checked every proof. More precisely, every numbered result of the present paper — including the perturbation analysis of §4 — has been formally verified in Lean 4 using the Mathlib library, with no `sorry`s, no custom axioms, and no `native_decide` or `unsafe` blocks. A line-by-line correspondence between the manuscript and the formal proofs is provided in Appendix A.

1.2 Markov decision processes and the Bellman equation

The game is a finite-state *Markov decision process* (MDP): the state is the number m of coins remaining, the action is the number $i \in \{1, \dots, k\}$ of heads to set aside (where k is the random number of heads), and the transition is $m \mapsto m - i$. There are two absorbing states: *win* ($m = 0$, all coins collected) and *lose* (reached when $k = 0$ in some round). A *policy* (or *strategy*) π specifies, for each state n and each observed number $k \in \{1, \dots, n\}$ of heads, the number $i_\pi(n, k) \in \{1, \dots, k\}$ of heads to set aside; we write v_n^π for the resulting winning probability starting from state n.

Because every policy absorbs with probability at least $p^m + (1-p)^m > 0$ per round, the MDP is a *stochastic shortest-path* (SSP) problem in the sense of Bertsekas and Tsitsiklis [2]: the Bellman operator is a contraction and the optimal value function is its unique fixed point.

Definition 1.1 (Optimal winning probability)**.** Let $w_{0,p} := 1$. For $n \geq 1$ define

$$w_{n,p} \;:=\; p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \max_{j \leq m \leq n-1} w_{m,p}, \tag{1}$$

where $q = 1 - p$ and the empty sum (at $n = 1$) gives $w_{1,p} = p$.

The term p^n accounts for the event that all n coins land heads (an immediate win). Each summand with $j \geq 1$ accounts for the event that exactly j coins land tails; the player then chooses to keep some number $m \in \{j, \dots, n-1\}$ of coins (setting aside $n - m$ heads), and the game continues from state m with optimal value $w_{m,p}$.

The Bellman equation (1) is the standard optimality equation for this SSP–MDP. For general background on Markov decision processes and the Bellman equation we refer to the monographs of Puterman [5], Bertsekas [1], and Stokey, Lucas and Prescott [6].

1.3 Strategies One and All

Under strategy One the player always keeps $m = n - 1$ coins (except when $k = n$), so the Bellman recursion collapses to

$$a_{n,p} \;=\; p^n + \big(1 - p^n - (1-p)^n\big)\, a_{n-1,p}, \qquad a_{0,p} := 1. \tag{2}$$

Under strategy All the player always keeps $m = j$ coins (the number of tails), giving

$$b_{n,p} \;=\; p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j\, b_{j,p}, \qquad b_{0,p} := 1. \tag{3}$$

Note that the recursions for both a and b are *linear* (no max operator), which makes them analytically more tractable than the Bellman equation (1).

1.4 Outline

In Section 2 we prove that at $p = \frac{1}{2}$ the optimal winning probability is $w_{n,1/2} = \frac{1}{2}$ for every n, and that in fact every reasonable strategy achieves this value. In Section 3 we show that for $p > \frac{1}{2}$ the strategy ONE is always optimal and $n \mapsto w_{n,p}$ is strictly increasing; as a consequence (§3.1) the limit $W(p) = \lim_n w_{n,p}$ exists and admits a closed-form series representation. Section 4 treats the case $p < \frac{1}{2}$ near $\frac{1}{2}$: we introduce a first-order perturbation in $\delta = \frac{1}{2} - p$, derive a recursion for the leading coefficient c_n, prove that it simplifies to a linear recursion for $n \geq 7$, and compute the limit $L = \lim c_n$. Section 5 complements the first-order analysis with numerical experiments that illustrate the shape of $n \mapsto w_{n,p}$ away from $p = \frac{1}{2}$, including the emergence of local maxima for p bounded away from $\frac{1}{2}$ (invisible in the first-order expansion), and Section 6 collects concluding observations and open questions. Appendix A describes the formal verification of all results in Lean 4 and provides a line-level map between the manuscript and the formalization on GitHub.

2 The fair coin: $p = \frac{1}{2}$

Theorem 2.1. *For every $n \geq 1$, $w_{n,1/2} = \frac{1}{2}$. More generally, for any policy π satisfying $i_\pi(n,n) = n$ for every n (i.e. the player accepts the immediate win when all coins land heads), the winning probability is $v_n^\pi = \frac{1}{2}$.*

In particular, both strategies ONE and ALL achieve the optimum: $a_{n,1/2} = b_{n,1/2} = w_{n,1/2} = \frac{1}{2}$.

Proof. We use the binomial identity

$$\sum_{j=1}^{n-1} \binom{n}{j} \left(\frac{1}{2}\right)^n = 1 - 2^{1-n}. \tag{4}$$

Step 1. We show $w_{n,1/2} = \frac{1}{2}$ by strong induction. For $n = 1$: $w_{1,1/2} = \frac{1}{2}$. For $n \geq 2$, suppose $w_{m,1/2} = \frac{1}{2}$ for all $1 \leq m \leq n-1$. Then every suffix-maximum equals $\frac{1}{2}$, so

$$\begin{aligned} w_{n,1/2} &= 2^{-n} + \frac{1}{2} \sum_{j=1}^{n-1} \binom{n}{j} 2^{-n} \\ &= 2^{-n} + \tfrac{1}{2}\big(1 - 2^{1-n}\big) \\ &= 2^{-n} + \tfrac{1}{2} - 2^{-n} \;=\; \tfrac{1}{2}. \end{aligned}$$

Step 2. For a general policy π with $i_\pi(n,n) = n$, conditioning on the number of heads $K \sim \mathrm{Bin}(n, \frac{1}{2})$ gives

$$v_n^\pi = 2^{-n} \cdot 1 + \sum_{k=1}^{n-1} \binom{n}{k} 2^{-n}\, v^\pi_{n - i_\pi(n,k)}.$$

Since $1 \le i_\pi(n,k) \le k \le n-1$ for $k < n$, the subscript $n - i_\pi(n,k)$ lies in $\{1,\dots,n-1\}$, and by strong induction each $v^\pi_{n-i_\pi(n,k)} = \frac{1}{2}$. The same calculation as in Step 1 gives $v^\pi_n = \frac{1}{2}$. $\square$

Remark 2.2. A policy that refuses the immediate win at $k = n$ (i.e. $i_\pi(n,n) < n$) achieves strictly less than $\frac{1}{2}$: the contribution of the $k = n$ event drops from 2^{-n} to $2^{-n} \cdot v^\pi_{n-i_\pi(n,n)} < 2^{-n}$, while all other terms remain at most $\frac{1}{2} \cdot \binom{n}{k} 2^{-n}$, giving $v^\pi_n < \frac{1}{2}$.

3 Above the fair coin: $p > \frac{1}{2}$

Theorem 3.1. *For $p > \frac{1}{2}$ and all $n \ge 2$:*

(i) $w_{n,p} > w_{n-1,p}$, *i.e. $n \mapsto w_{n,p}$ is strictly increasing.*

(ii) $w_{n-1,p} < \dfrac{p^n}{p^n + q^n}$.

(iii) *Strategy* One *is optimal: $w_{n,p} = a_{n,p}$.*

Proof. Note that $n \mapsto \dfrac{p^n}{p^n + q^n} = \dfrac{1}{1 + (q/p)^n}$ is increasing for $p \in (\frac{1}{2}, 1)$. We prove (i) and (ii) simultaneously by strong induction on n.

Base case $n = 2$. One computes $w_{1,p} = p$ and $w_{2,p} = p^2 + (1 - p^2 - q^2)p = 3p^2 - 2p^3$. Then $w_{2,p} - w_{1,p} = p(1-p)(2p-1) > 0$ since $p > \frac{1}{2}$. Also $w_{1,p} = p = p/(p+q) < p^2/(p^2+q^2)$.

Inductive step. Assume (i) and (ii) hold for all indices up to $n-1$. Since $n \mapsto w_{n,p}$ is increasing on $\{1,\dots,n-1\}$ by hypothesis, the suffix-maximum $\max_{j \le m \le n-1} w_{m,p} = w_{n-1,p}$ for every $j \in \{1,\dots,n-1\}$. Hence the Bellman equation (1) reduces to

$$w_{n,p} = p^n + (1 - p^n - q^n)\, w_{n-1,p},$$

which is exactly the recursion (2) for strategy One. This proves (iii).

For (i), $w_{n,p} > w_{n-1,p}$ iff $p^n(1 - w_{n-1,p}) > q^n w_{n-1,p}$, i.e. $w_{n-1,p} < p^n/(p^n + q^n)$, which is hypothesis (ii) at level $n-1$.

For (ii) at level n, we need $w_{n,p} < p^{n+1}/(p^{n+1} + q^{n+1})$. Using the recursion and the inductive bound on $w_{n-1,p}$, a direct computation shows this reduces to the inequality $(q/p)^n < (q/p)^{n-1}$, which holds since $q/p < 1$. $\square$

3.1 The limit $W(p) = \lim_{n\to\infty} w_{n,p}$

The collapse of the Bellman suffix-maximum, which is the central content of Theorem 3.1, has a useful by-product: the Bellman recursion for $w_{n,p}$ is genuinely *linear* for every $n \ge 1$.

Corollary 3.2 (Linear recursion above $\frac{1}{2}$)**.** *For $p > \frac{1}{2}$ and every $n \ge 1$,*

$$w_{n,p} \;=\; p^n + \big(1 - p^n - q^n\big)\, w_{n-1,p}, \tag{5}$$

with $w_{0,p} = 1$.

Proof. For $n = 1$ the sum in (1) is empty and $w_{1,p} = p = p^1 + (1 - p - q) \cdot 1$ since $p + q = 1$. For $n \geq 2$, combine the suffix-max collapse from the proof of Theorem 3.1 with the binomial identity $\sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j = 1 - p^n - q^n$. □

The linear recursion (5) is the analogue, for $p > \frac{1}{2}$, of the linear c_n-recursion of Proposition 4.8 below. It is structurally *simpler*: it holds for every $n \geq 1$ (no preliminary joint-induction step is needed), the coefficients are explicit polynomials in p, and the bounded monotone sequence converges trivially by the Weierstrass criterion.

Theorem 3.3 (Limit and explicit formula above $\frac{1}{2}$)**.** *For every $p \in (\frac{1}{2}, 1)$ the limit $W(p) := \lim_{n\to\infty} w_{n,p}$ exists, satisfies $p \leq W(p) < 1$, and admits the closed-form representation*

$$W(p) \;=\; \sum_{k=1}^{\infty} p^k \prod_{j=k+1}^{\infty} \big(1 - p^j - q^j\big). \tag{6}$$

The series converges absolutely at geometric rate.

Proof. Existence follows from Theorem 3.1 (i) (monotone, bounded above by 1). The lower bound $W(p) \geq w_{1,p} = p$ is immediate from monotonicity; combined with $p > \frac{1}{2}$ this gives $W(p) > \frac{1}{2}$.

For the explicit formula, iterate (5) from $k = 1$ up to $k = n$:

$$w_{n,p} \;=\; \underbrace{\prod_{k=1}^{n} \big(1 - p^k - q^k\big) \cdot w_{0,p}}_{=0} \;+\; \sum_{k=1}^{n} p^k \prod_{j=k+1}^{n} \big(1 - p^j - q^j\big), \tag{7}$$

where the first product vanishes because its $k = 1$ factor is $1 - p - q = 0$. For $j \geq 2$ the factors $1 - p^j - q^j = \sum_{i=1}^{j-1} \binom{j}{i} p^{j-i} q^i$ lie in $(0, 1)$ (positive from the $i = 1$ term; strictly less than 1 since we subtracted $p^j + q^j > 0$). Together with $\sum_{j\geq 2}(p^j + q^j) < \infty$, the partial products $\prod_{j=k+1}^{n}(1 - p^j - q^j)$ converge as $n \to \infty$ to a strictly positive limit $\Pi_k \in (0, 1)$, and the resulting series $\sum_k p^k \Pi_k$ converges absolutely by comparison with $\sum_k p^k$. By Tannery's theorem (cf. Knopp [4, §41]) with dominator p^k, limit and summation commute, giving (6).

It remains to show $W(p) < 1$. Set $U(p) := 1 - W(p)$ and $u_n := 1 - w_{n,p}$; subtracting (5) from 1 and using $p^n + q^n + (1 - p^n - q^n) = 1$ gives the companion recursion $u_n = q^n + (1 - p^n - q^n) u_{n-1}$ with $u_0 = 0$. Iterating identically to (7) gives the analogue

$$U(p) \;=\; \sum_{k=1}^{\infty} q^k \Pi_k,$$

which is a strictly positive sum (every term is positive: $q > 0$ and $\Pi_k > 0$). Hence $W(p) = 1 - U(p) < 1$. □

Remark 3.4 (Parallel with Theorem 4.10). The iterated identity (7) and its limit formula (6) have the same structure as the finite iteration (17) and the formula (15) for L below: in both cases a linear recursion of the form $x_n = A_n + (1 - B_n)\, x_{n-1}$ is iterated and Tannery's theorem identifies the resulting limit as $\lim x_{n_0-1} \prod(1-B_j) + \sum A_k \prod(1-B_j)$, with dominator

$\sum |A_k| < \infty$. The two differ only in the coefficients: for $W(p)$, $A_k = p^k$ and $B_k = p^k + q^k$; for L, $A_k = A_{\text{lin}}(k)$ and $B_k = B_{\text{lin}}(k)$ from Proposition 4.8, and the structural step leading to the linear recursion is trivial above $\frac{1}{2}$ (immediate from monotonicity) but requires the joint induction of §4.3 below.

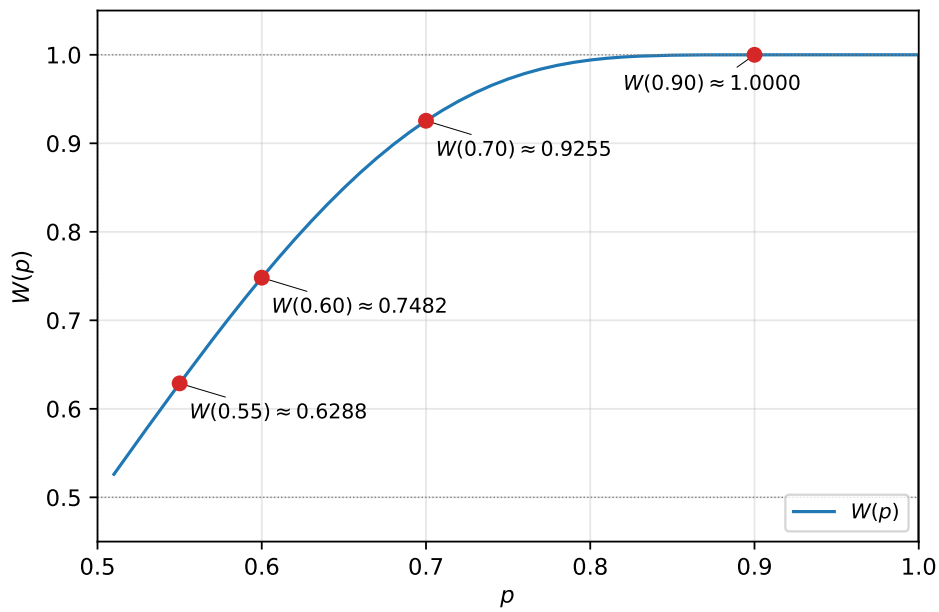

Figure 1: The limit $W(p) = \lim_{n\to\infty} w_{n,p}$ as a function of p, for $p \in (\frac{1}{2}, 1)$. Computed via the recursion of Corollary 3.2 iterated to $n = 400$ in 80-digit arithmetic. Selected values: $W(0.55) \approx 0.6288$, $W(0.6) \approx 0.7482$, $W(0.7) \approx 0.9255$, $W(0.9) \approx 0.99998$. The limit is monotone in p, bounded below by the identity line (since $W(p) \geq p$) and strictly below 1 by Theorem 3.3.

4 Below the fair coin: perturbation in $\delta = \frac{1}{2} - p$

For $p < \frac{1}{2}$ the structure of $n \mapsto w_{n,p}$ is far richer: numerical experiments show that the sequence develops local minima and maxima, and the optimal strategy is neither One nor All in general; see [7], Section 10, and Section 5 of the present paper for illustrations.

To get analytical traction, we study the regime $p = \frac{1}{2} - \delta$ with $\delta > 0$ small.

4.1 The deficit and its recursion

Definition 4.1. For $p \in (0, 1)$ and $n \geq 1$, define the *deficit* $\Delta_{n,p} := \frac{1}{2} - w_{n,p}$.

By Theorem 2.1, $\Delta_{n,1/2} = 0$ for every $n \geq 1$. At $n = 0$ the convention $w_{0,p} := 1$ ("no coins left" is a trivial win) yields $\Delta_{0,p} = -\frac{1}{2}$ for every p; this value will reappear in Proposition 4.2.

Proposition 4.2. *The deficit satisfies*

$$\Delta_{n,p} \;=\; \frac{q^n - p^n}{2} + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \min_{j \le m \le n-1} \Delta_{m,p}, \tag{8}$$

with $\Delta_{0,p} = -\frac{1}{2}$. *In particular:*

(i) *For* $p < \frac{1}{2}$: $\Delta_{n,p} > 0$ *for every* $n \ge 1$ *(i.e.* $w_{n,p} < \frac{1}{2}$*).*

(ii) *For* $p > \frac{1}{2}$: $\Delta_{n,p} < 0$ *for every* $n \ge 1$ *(i.e.* $w_{n,p} > \frac{1}{2}$*).*

Proof. Substitute $w_{m,p} = \frac{1}{2} - \Delta_{m,p}$ into the Bellman equation (1). The max on w_m becomes a min on Δ_m. The constant terms $\frac{1}{2} \cdot \big[\text{binomial weights}\big]$ cancel by the identity $p^n + \sum_{j=1}^{n-1} \binom{n}{j} p^{n-j} q^j \cdot \frac{1}{2} = \frac{1}{2} - \frac{1}{2}(q^n - p^n)$, leaving (8). The sign claims follow by induction using $q^n - p^n > 0$ (resp. < 0) and the positivity (resp. negativity) of the min terms. □

4.2 First-order expansion

Set $p = \frac{1}{2} - \delta$ with $\delta > 0$ small. Then $q = \frac{1}{2} + \delta$, and the constant term in (8) is

$$\frac{q^n - p^n}{2} = \frac{(\frac{1}{2} + \delta)^n - (\frac{1}{2} - \delta)^n}{2} = \frac{n\delta}{2^{n-1}} + O(\delta^3).$$

The binomial weights become $\binom{n}{j} 2^{-n} + O(\delta)$, and the min terms are $O(\delta)$ by Proposition 4.2. Thus $\Delta_{n,p} = O(\delta)$.

Proposition 4.3 (First-order coefficient)**.** *Write* $\Delta_{n,p} = c_n \delta + O(\delta^2)$ *as* $\delta \to 0^+$. *Then* $c_1 = 1$ *and for* $n \ge 2$*:*

$$c_n \;=\; \frac{n}{2^{n-1}} + \frac{1}{2^n} \sum_{j=1}^{n-1} \binom{n}{j} \min_{j \le m \le n-1} c_m. \tag{9}$$

Proof. Match first-order coefficients in (8). The constant term contributes $n/2^{n-1}$. Each binomial weight is $\binom{n}{j} 2^{-n} + O(\delta)$; the min of the $\Delta_m = c_m \delta + O(\delta^2)$ at first order is $(\min c_m)\delta + O(\delta^2)$. Products of $O(\delta)$ terms are $O(\delta^2)$. □

Example 4.4. The first few values of c_n:

$$c_1 = 1, \quad c_2 = \tfrac{3}{2}, \quad c_3 = \tfrac{27}{16} = 1.6875, \quad c_4 = \tfrac{111}{64} \approx 1.7343,$$
$$c_5 = \tfrac{3555}{2048} \approx 1.7358, \quad c_6 = \tfrac{113337}{65536} \approx 1.7293.$$

The sequence increases on $\{1, \ldots, 5\}$ and decreases on $\{5, 6, 7, ...\}$ (as we will prove below). In particular $c_5 > c_4 > c_6$. Each of the six identities above is formally verified in Lean 4 (`c_one`, ..., `c_six`); see Appendix A, Table 3, for direct links to the source. Figure 2 plots the sequence up to $n = 25$, together with the limit $L = \lim_{n\to\infty} c_n \approx 1.7035$ and the uniform lower bound $\frac{27}{16} = 1.6875$ established in Lemma 4.6.

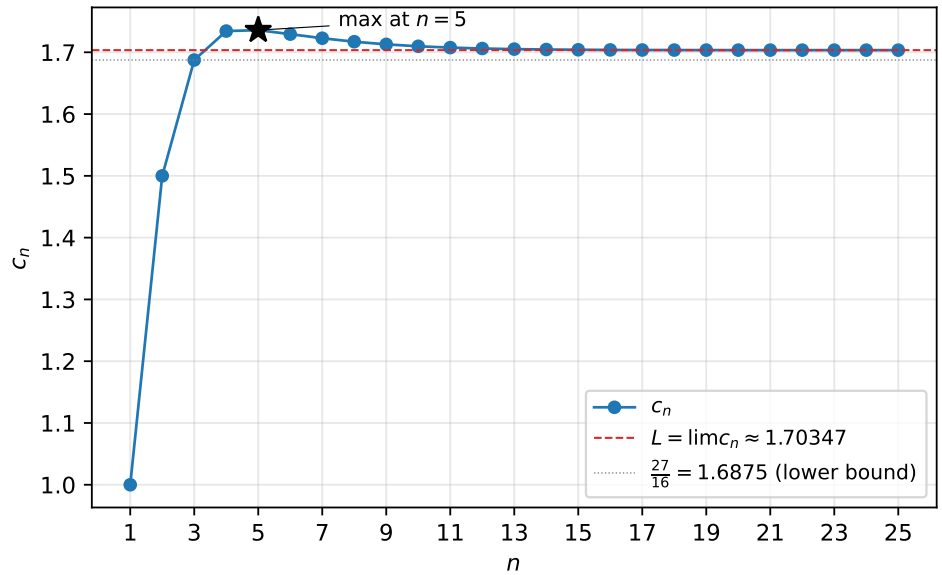

Figure 2: The sequence $n \mapsto c_n$ for $1 \le n \le 25$. Computed via the recursion of Proposition 4.3 in 80-digit rational arithmetic. The sequence increases on $\{1, \dots, 5\}$ (with maximum at $n = 5$, marked by a star), strictly decreases for $n \ge 5$, and converges from above to $L \approx 1.7035$. The lower bound $\frac{27}{16}$ from Lemma 4.6 is shown as a dotted line.

4.3 Collapse, lower bound, and linear recursion

The four results below — the asymptotic collapse of the min in (9) (Lemma 4.5), the uniform lower bound $c_n \ge \frac{27}{16}$ for $n \ge 4$ (Lemma 4.6), the strict decrease of $(c_n)_{n\ge5}$ (Lemma 4.7), and the linear recursion for $n \ge 7$ (Proposition 4.8) — are mutually entangled: each, taken alone, depends on parts of the others. Stating them in isolation produces an apparent circular dependency. We therefore state all four results first and then prove them simultaneously by induction on n, with the IH at level $< n$ supplying exactly what is needed at level n.

Lemma 4.5 (Asymptotic collapse)**.** *For all $n \ge 7$ and $j \in \{1, \dots, n-1\}$:*

$$\min_{j \le m \le n-1} c_m \;=\; \begin{cases} c_j & \text{if } j \in \{1,2,3\}, \\ c_{n-1} & \text{if } j \in \{4, \dots, n-1\}. \end{cases} \tag{10}$$

Lemma 4.6. *$c_n \ge \frac{27}{16}$ for every $n \ge 4$.*

Lemma 4.7. *The sequence $(c_n)_{n\ge5}$ is strictly decreasing.*

Proposition 4.8 (Linear recursion)**.** *For $n \ge 7$:*

$$c_n \;=\; A_n + (1 - B_n)\, c_{n-1}, \tag{11}$$

where

$$A_n := \frac{n}{2^{n-1}} + \frac{n\, c_1 + \binom{n}{2} c_2 + \binom{n}{3} c_3}{2^n}, \qquad B_n := \frac{2 + n + \binom{n}{2} + \binom{n}{3}}{2^n}. \tag{12}$$

Moreover the algebraic identity

$$A_n - \tfrac{27}{16}B_n \;=\; -\frac{3(n^2-15n+36)}{32\cdot 2^n} \tag{13}$$

holds for every $n \geq 1$ (independently of the linear recursion).

Joint proof of Lemmas 4.5, 4.6, 4.7 and Proposition 4.8. The identity (13) is a direct algebraic computation from (12) using $\binom{n}{2} = \frac{n(n-1)}{2}$ and $\binom{n}{3} = \frac{n(n-1)(n-2)}{6}$; we use it freely below. Formally, it is verified in Lean as the lemma `alg_id`; see Appendix A, Table 3.

Let $P(n)$ denote the conjunction of the following four claims (each understood vacuously when its index range is empty):

(a) Lemma 4.5 holds at level n (when $n \geq 7$);

(b) $c_n \geq \frac{27}{16}$ (when $n \geq 4$);

(c) $c_n < c_{n-1}$ (when $n \geq 6$);

(d) the linear recursion (11) holds at level n (when $n \geq 7$).

We prove $P(n)$ for all $n \geq 4$ by strong induction on n.

Base cases $n \in \{4,5,6\}$. By Example 4.4, $c_4 = \frac{111}{64}$, $c_5 = \frac{3555}{2048}$, $c_6 = \frac{113337}{65536}$, all of which exceed $\frac{27}{16} = \frac{108}{64}$, so (b) holds. Also $c_5 > c_6$, so (c) holds at $n = 6$. Claims (a) and (d) are vacuous for $n \leq 6$.

Inductive step. Fix $n \geq 7$ and assume $P(m)$ for all $4 \leq m < n$. In particular: by IH(b), $c_m \geq \frac{27}{16}$ for $4 \leq m \leq n-1$; by IH(c), $c_5 > c_6 > \cdots > c_{n-1}$. We now establish (a)–(d) at level n.

(a) Collapse. For $j \in \{1,2,3\}$: from $c_1 = 1 < c_2 = \frac{3}{2} < c_3 = \frac{27}{16}$ (Example 4.4) and $c_m \geq \frac{27}{16} = c_3$ for $m \geq 4$ (IH(b)), the minimum over $\{j,\ldots,n-1\}$ is c_j.

For $j \in \{4,\ldots,n-1\}$: iterating IH(c) gives the chain $c_5 > c_6 > \cdots > c_{n-1}$, so in particular $c_m \geq c_{n-1}$ for every $m \in \{5,\ldots,n-1\}$. For $m = 4$, $c_4 = \frac{111}{64} > c_6$ (Example 4.4); using the segment $c_6 \geq c_{n-1}$ of the same chain, we obtain $c_4 \geq c_{n-1}$ as well. Hence the minimum over $\{j,\ldots,n-1\}$ for $j \geq 4$ is c_{n-1}.

(d) Linear recursion. Substituting the collapse (a) just proved into (9) and splitting the sum at $j = 3$ yields (11)–(12). Formally, this substitution is verified in Lean as the theorem `c_linear_rec`; see Appendix A, Table 3.

(b) Lower bound. From (d) and (13),

$$c_n - \tfrac{27}{16} \;=\; \big(A_n - \tfrac{27}{16}B_n\big) + (1-B_n)\big(c_{n-1} - \tfrac{27}{16}\big).$$

By IH(b), $c_{n-1} - \frac{27}{16} \geq 0$. Note also $0 < B_n < 1$ for all $n \geq 7$: the numerator $2+n+\binom{n}{2}+\binom{n}{3}$ in (12) is strictly less than $2^n = \sum_{j=0}^{n}\binom{n}{j}$, the difference being $\sum_{j=4}^{n-1}\binom{n}{j} > 0$. For $n \in \{7,\ldots,12\}$, the quadratic $n^2-15n+36$ is non-positive on $\{3,\ldots,12\}$ (its roots are 3 and 12), so $A_n - \frac{27}{16}B_n \geq 0$ by (13), and (b) follows. For $n \geq 13$ we argue cumulatively: define $\varepsilon_n := c_n - \frac{27}{16}$ and $\delta_n := 3(n^2-15n+36)/(32\cdot 2^n)$, so (d) gives $\varepsilon_n = (1-B_n)\varepsilon_{n-1} - \delta_n$

for every $n \geq 13$. Iterating this identity from level 12 upwards and using the trivial bound $0 < 1 - B_k \leq 1$ yields, for every $n \geq 13$,

$$\varepsilon_n \;=\; \prod_{k=13}^{n}(1-B_k)\,\varepsilon_{12} \;-\; \sum_{k=13}^{n}\Big(\prod_{j=k+1}^{n}(1-B_j)\Big)\delta_k \;\geq\; \Big(\prod_{k=13}^{n}(1-B_k)\Big)\varepsilon_{12} \;-\; \sum_{k=13}^{n}\delta_k. \tag{14}$$

The starting value $\varepsilon_{12} > \frac{1}{60}$ is verified by exact rational computation (formally: `c_twelve_buffer`); the cumulative negative contribution is bounded by $\sum_{n\geq 13}\delta_n \leq \frac{297}{65536} < \frac{1}{200}$ (using $\sum_{n\geq 13} n^2/2^n = \frac{99}{2048}$; formally: `delta_tail_bound`); meanwhile $\prod_{n\geq 13}(1-B_n) > \frac{7}{8}$ (using $\sum_{n\geq 13} B_n < \frac{1}{8}$; formally: `B_tail_bound`). Inserting these three inequalities into (14) gives $\varepsilon_n > \frac{1}{60}\cdot\frac{7}{8} - \frac{1}{200} = \frac{23}{2400} > 0$ for every $n \geq 13$. The iterated inequality itself is formally verified (in the slightly weaker Bernoulli-style form $\prod(1-B_k) \geq 1-\sum B_k$) as `cum_eps_bound`.

(*Note for an inductive treatment:* the sequence $(\delta_n)_{n\geq 13}$ is not monotone — one computes $\delta_{14}/\delta_{13} = \frac{11}{10}$ — but its ratio decreases through 1 at $n = 14$ and tends to $\frac{1}{2}$. An iterated proof of the bound on $\sum_{n\geq 13}\delta_n$ should therefore split at $n = 14$; the closed-form route above sidesteps this subtlety.)

(c) Strict decrease. By (d), $c_n - c_{n-1} = A_n - B_n c_{n-1}$. Using $c_{n-1} \geq \frac{27}{16}$ (IH(b)) and $B_n > 0$, this is $\leq A_n - \frac{27}{16}B_n < 0$ for $n \geq 13$ by (13) ($n^2 - 15n + 36 > 0$ there). For $n \in \{7, \ldots, 12\}$, $c_n < c_{n-1}$ is verified by direct rational computation (chain via the linear recursion from c_6 up to c_n, then numerical comparison). This completes the inductive step. The combined statement of (c) for all $n \geq 5$ is exposed as `c_strict_anti_from_five`; the corresponding statement of (b) for all $m \geq 4$ as `c_ge_27_16_full`. □

4.4 The limit

Since $(c_n)_{n\geq 5}$ is strictly decreasing (Lemma 4.7) and bounded below by $\frac{27}{16}$ (Lemma 4.6), the Weierstraß monotone-convergence criterion gives at once:

Corollary 4.9 (Existence of the limit)**.** *The limit $L := \lim_{n\to\infty} c_n$ exists and satisfies $\frac{27}{16} \leq L \leq c_5 = \frac{3555}{2048}$.*

Formally verified as `c_limit_exists`. The actual mathematical content of this subsection is the *explicit* formula for L:

Theorem 4.10 (Explicit formula for the limit)**.** *For any $n_0 \geq 7$,*

$$L \;=\; c_{n_0-1}\prod_{m\geq n_0}(1-B_m) \;+\; \sum_{k\geq n_0} A_k \prod_{m>k}(1-B_m). \tag{15}$$

Both the infinite product and the series converge at geometric rate since $A_n, B_n = O(n^3/2^n)$. Numerically,

$$L \;=\; 1.70347176087173673645\ldots \tag{16}$$

Proof. Fix $n_0 \geq 7$. Iterating the linear recursion (11) from n_0 up to n yields, for every $n \geq n_0$,

$$c_n \;=\; c_{n_0-1}\prod_{m=n_0}^{n}(1-B_m) \;+\; \sum_{k=n_0}^{n} A_k \prod_{m=k+1}^{n}(1-B_m). \tag{17}$$

By Corollary 4.9 we already know that the left-hand side converges to L; the content of the theorem is that the right-hand side converges to the asserted explicit expression. We pass to the limit $n \to \infty$ in each summand.

Step 1 (infinite product). Since $B_m = O(m^3/2^m)$, the series $\sum_{m \geq n_0} B_m$ converges absolutely. Because $0 < B_m < 1$ for all $m \geq 7$ (shown in the inductive proof of the previous subsection), the partial products $\prod_{m=n_0}^{n}(1 - B_m)$ converge as $n \to \infty$ to a strictly positive limit $\prod_{m \geq n_0}(1 - B_m)$; cf. Knopp [4, §29].

Step 2 (series of products). For each fixed $k \geq n_0$ the same argument gives $\prod_{m=k+1}^{n}(1 - B_m) \to \prod_{m>k}(1 - B_m)$ as $n \to \infty$. Moreover every factor lies in $(0, 1)$, so each finite product lies in $(0, 1]$, and

$$\left| A_k \prod_{m=k+1}^{n} (1 - B_m) \right| \;\leq\; |A_k| \qquad (n \geq k). \tag{18}$$

Since $A_k = O(k^3/2^k)$, the series $\sum_{k \geq n_0} |A_k|$ converges. The bound (18) therefore supplies a summable dominant, so by Tannery's theorem — the dominated convergence theorem for series, see Knopp [4, §41] — limit and summation commute:

$$\lim_{n \to \infty} \sum_{k=n_0}^{n} A_k \prod_{m=k+1}^{n} (1 - B_m) \;=\; \sum_{k \geq n_0} A_k \prod_{m>k} (1 - B_m).$$

Conclusion. Taking $n \to \infty$ in (17) and using Steps 1 and 2 gives (15). Both the product and the series converge at geometric rate since $A_n, B_n = O(n^3/2^n)$. □

4.5 Shape of the optimal-value sequence near $p = \frac{1}{2}$

Corollary 4.11. *For $\delta > 0$ sufficiently small and $p = \frac{1}{2} - \delta$:*

(i) $w_{n-1,p} - w_{n,p} = (c_n - c_{n-1})\delta + O(\delta^2)$.

(ii) *The sequence $n \mapsto w_{n,p}$ has, to first order in δ, a strict local minimum at $n = 5$ (since $c_5 > c_4$ and $c_5 > c_6$).*

(iii) *There is no local maximum at first order: c_n is eventually decreasing, so $w_{n,p}$ is eventually increasing.*

Remark 4.12. Numerical computation shows that the sequence $n \mapsto w_{n,p}$ does develop strict local maxima for p bounded away from $\frac{1}{2}$ (e.g. at $n = 9$ for $p = 0.42$, verified by exact rational arithmetic). This is consistent with the corollary: local maxima are a *non-perturbative* phenomenon invisible to the first-order expansion. Van Doorn [7], Section 10, conjectures that the optimal strategy for $p < \frac{1}{2}$ is "infinitely complex."

5 Numerical experiments for $p < \frac{1}{2}$

The first-order analysis of Section 4 describes the shape of $n \mapsto w_{n,p}$ only in a shrinking neighbourhood of $p = \frac{1}{2}$. Corollary 4.11 guarantees, for every sufficiently small $\delta > 0$, a unique local minimum at $n = 5$ and no local maximum. Numerical computation of $w_{n,p}$

confirms this picture for p close to $\frac{1}{2}$, and shows that local maxima emerge for p bounded away from $\frac{1}{2}$.

All values of $w_{n,p}$ below are computed by the exact Bellman recursion (1), evaluated in arbitrary-precision rational (or 80-decimal-digit) arithmetic via `mpmath`. The code is available in the `simulation/` subdirectory of the repository.

Figure 3 plots $w_{n,p}$ against n for five values of p. For $p = 0.49$ (first-order regime) the sequence bottoms out at $n = 5$ and increases thereafter, in agreement with Corollary 4.11. For $p = 0.42$ and $p = 0.45$ a non-monotone structure emerges: the sequence first decreases, passes through a local minimum, then rises to a local *maximum* before decreasing again and converging.

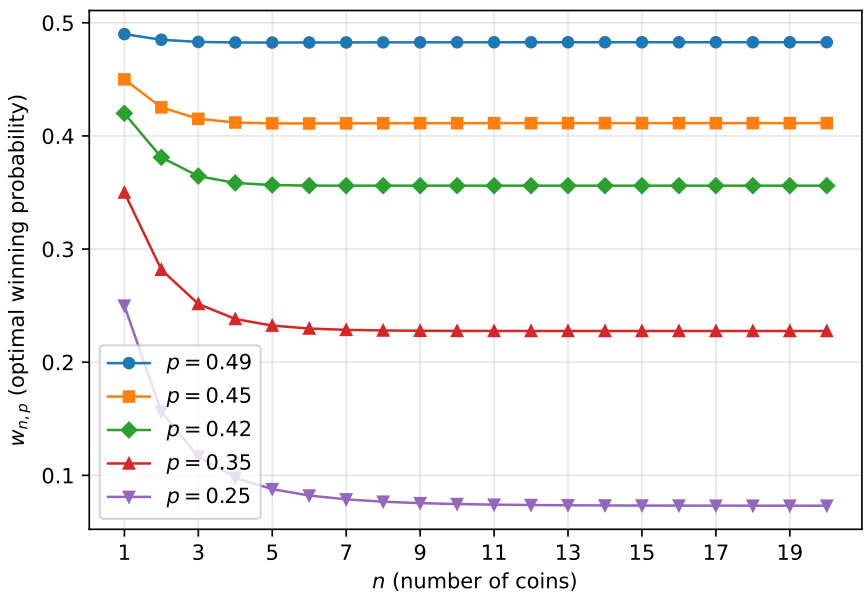

Figure 3: The optimal winning probability $w_{n,p}$ as a function of n for five values of $p < \frac{1}{2}$. Computed by the exact Bellman recursion in 80-digit arithmetic. For $p = 0.49$ only the predicted local minimum at $n = 5$ is visible; for $p = 0.42$ and $p = 0.45$ local maxima appear. For $p = 0.35$ and $p = 0.25$ no local extrema occur within $n \leq 20$.

The non-monotone behaviour is invisible in the plot at the scale of Figure 3, because the oscillations are small compared with the total variation of $w_{n,p}$ in n. Figure 4 zooms in on $p = 0.42$ and $p = 0.45$ and highlights the local minima and maxima found by direct inspection of the sequence.

Table 1 lists the numerical values of $w_{n,p}$ for $1 \leq n \leq 20$ and the five values of p considered in Figure 3. The local extrema are tabulated in Table 2.

6 Discussion

For $p > \frac{1}{2}$, the situation is much simpler than for $p < \frac{1}{2}$: the linear recursion of Corollary 3.2 holds for every $n \geq 1$, and the limit $W(p) = \lim_{n\to\infty} w_{n,p}$ admits the closed form (6);

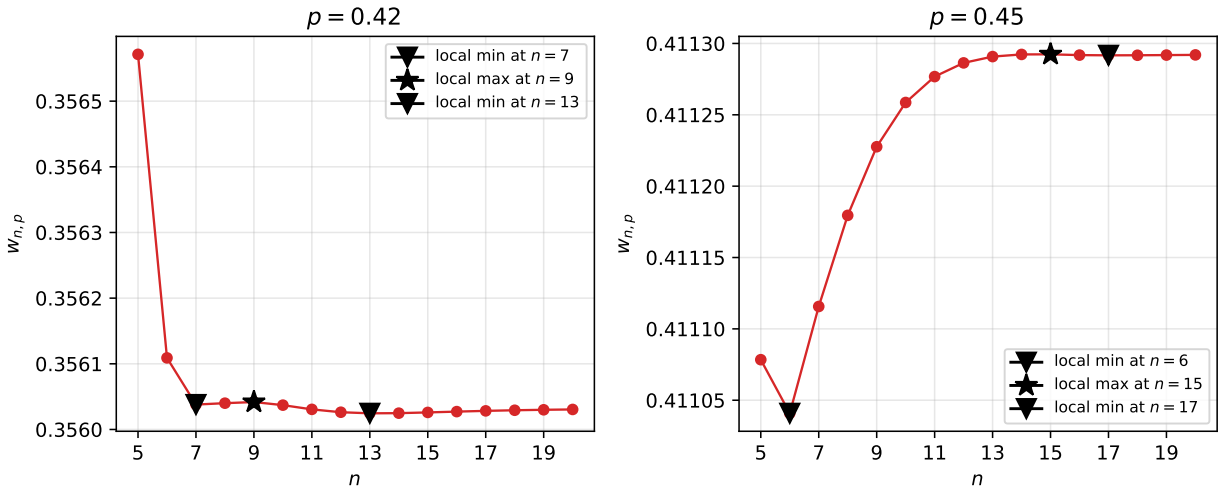

Figure 4: Zoomed-in view of $w_{n,p}$ for $p \in \{0.42, 0.45\}$ and $n \in [5, 20]$. Stars mark local maxima, inverted triangles mark local minima. At $p = 0.42$ there is a local maximum at $n = 9$; at $p = 0.45$ there is a local maximum at $n = 15$.

numerically, $W(p)$ is a smooth, strictly increasing function of $p \in (\frac{1}{2}, 1)$ with $p \leq W(p) < 1$ (Figure 1). The bulk of this paper is therefore concerned with the more delicate regime $p < \frac{1}{2}$, where two features of the numerical data in §5 deserve further comment.

(a) Local maxima are a non-perturbative phenomenon. Corollary 4.11(iii) rules out local maxima of $n \mapsto w_{n,p}$ *to first order in* $\delta = \frac{1}{2} - p$: for any fixed n, there is a $\delta_0(n) > 0$ below which the predicted sign of $c_{n+1} - c_n < 0$ forces $w_{n,p} < w_{n+1,p}$. The local maximum at $n = 9$ for $p = 0.42$ ($\delta = 0.08$), and the one at $n = 15$ for $p = 0.45$ ($\delta = 0.05$), correspond to the threshold $\delta_0(n)$ being *larger* than the actual perturbation size at that n, so higher-order terms dominate. Numerically, $\delta_0(n)$ appears to decay rapidly in n (we see no local maxima at all for $p \leq 0.35$ within $n \leq 20$), which is consistent with the expected geometric decay of the first-order constants A_n and $B_n = O(n^3/2^n)$.

(b) The first-order picture holds robustly near $p = \frac{1}{2}$. For $p = 0.49$ ($\delta = 0.01$), the only local extremum within $n \leq 20$ is the predicted minimum at $n = 5$, with no spurious oscillations. This confirms numerically that the first-order range-of-validity extends at least to $\delta = 0.01$. Pinning down the size of $\delta_0(n)$ as a function of n remains open; we conjecture that it decays roughly like C/n^2 for some absolute constant C.

Remark 6.1. The picture for $p \leq 0.35$ in our data — a strictly decreasing sequence with no local extrema within $n \leq 20$ — is consistent with, but does not prove, monotone decrease on all of $\mathbb{N}$. Van Doorn [7] proves that strategy ALL is *not* optimal for $p < \frac{1}{2}$ (there exists n with $w_{n,p} \neq b_{n,p}$); equivalently, the sequence $n \mapsto w_{n,p}$ is not monotone non-increasing. The failure of monotonicity is, however, not always detectable as a local extremum in n; it can manifest as strict inequality $w_{n,p} > b_{n,p}$ without the sequence $w_{n,p}$ itself having oscillations. Investigating this gap is beyond the scope of the present paper.

n	$p = 0.49$	$p = 0.45$	$p = 0.42$	$p = 0.35$	$p = 0.25$
1	0.49000000	0.45000000	0.42000000	0.35000000	0.25000000
2	0.48500200	0.42525000	0.38102400	0.28175000	0.15625000
3	0.48309103	0.41514272	0.36450348	0.25147259	0.11669922
4	0.48258194	0.41184019	0.35845021	0.23820137	0.09781647
5	**0.48254059**	0.41107840	0.35657129	0.23236999	0.08782906
6	0.48259223	**0.41104082**	0.35610901	0.22977285	0.08215652
7	0.48264193	0.41111568	**0.35603793**	0.22859504	0.07878468
8	0.48268806	0.41117954	0.35604004	0.22805232	0.07672145
9	0.48272442	0.41122762	*0.35604176*	0.22779987	0.07543431
10	0.48275111	0.41125863	0.35603705	0.22768230	0.07462024
11	0.48276987	0.41127677	0.35603062	0.22762785	0.07410012
12	0.48278262	0.41128638	0.35602626	0.22760288	0.07376520
13	0.48279104	0.41129079	**0.35602460**	0.22759155	0.07354821
14	0.48279647	0.41129231	0.35602484	0.22758643	0.07340694
15	0.48279990	*0.41129237*	0.35602596	0.22758409	0.07331459
16	0.48280201	0.41129184	0.35602720	0.22758299	0.07325404
17	0.48280329	**0.41129168**	0.35602833	0.22758245	0.07321422
18	0.48280406	0.41129173	0.35602926	0.22758218	0.07318798
19	0.48280451	0.41129185	0.35602997	0.22758204	0.07317065
20	0.48280478	0.41129198	0.35603049	0.22758197	0.07315919

Table 1: Numerical values of $w_{n,p}$ for $n \in [1, 20]$ and five values of $p < \frac{1}{2}$. Local minima are marked in **bold**, local maxima in *italics*.

A Formal verification in Lean 4

All results of this paper have been formally verified in Lean 4 (toolchain `leanprover/lean4:v4.29.1`) using the Mathlib library at release tag `v4.29.1` (revision `5e932f97`). The formalization is publicly available at

`https://github.com/pfaffelh/coins`

and builds successfully against this Mathlib version with no open goals. The proofs use only the three standard Lean foundational axioms (`propext`, `Classical.choice`, `Quot.sound`); no additional axioms, `sorry`s, `native_decide`, `unsafe` blocks, or `opaque` definitions appear.

Table 3 lists every numbered theorem, proposition, lemma, and corollary stated in the main text together with its corresponding Lean theorem. Each hyperlink points to the exact line of the current pinned revision (commit `c60bcd3`).

The dominated-convergence step in the proof of Theorem 4.10 (Step 2) is discharged in Lean by Mathlib's `tendsto_tsum_of_dominated_convergence` (Tannery's theorem), and the infinite-product convergence in Step 1 uses `Real.multipliable_one_add_of_summable`. These are exactly the two analytic ingredients cited in the proof of Theorem 4.10.

A self-contained one-page summary of the headline objects, with their source-level definitions, full types, and axiom dependencies, is provided by the file `CoinsLean/Summary.lean`. Running

p	Local minima at $n =$	Local maxima at $n =$
0.49	5	—
0.45	6, 17	15
0.42	7, 13	9
0.35	—	—
0.25	—	—

Table 2: Local extrema of $n \mapsto w_{n,p}$ for $n \in [2, 19]$.

```
lake env lean CoinsLean/Summary.lean
```

from within the `CoinsLean/` directory prints the types of `w`, `deficit`, `c`, the main theorems of the paper, and the axiom list — verifying, via Lean's kernel, that the formalization type-checks as claimed.

Development history. The formalization was produced in two stages of interactive work with Anthropic's Claude. An earlier session using Claude Opus 4.6 established the initial scaffolding of the project — the Mathlib integration, the strategy-ALL file `Bellman.lean`, and the fair-coin computation in `HalfP.lean` (totalling about 100 lines). The present paper's formalization — the optimal-value definition `Optimal.lean`, the strategies module, the full proof of Theorem 3.1 in `Above.lean`, the limit analysis above $\frac{1}{2}$ in `AboveLimit.lean`, and the entire §4 perturbation module `Perturbation.lean` (about 3,700 lines of Lean total) — was produced in a subsequent sustained interaction with Claude Opus 4.7, with the author as mathematical guide and reviewer. Each step was decided through back-and-forth: the author selected which result to tackle next and acted as reviewer; Claude proposed, drafted, debugged, and revised proofs until they closed with `lake build` green and no `sorry`. A full prompt-by-prompt log of the second session is preserved in `claude.md`, and a thematic commentary tied to each commit is in `journal.md`.

Acknowledgements

The author thanks Joachim Breitner for introducing him to the game as well as help in Lean 4, Wouter van Doorn for valuable discussion, and Harald Binder for suggesting this new kind of doing mathematics. This research was supported by the DFG Project-ID 499552394–SFB 1597.

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Manuscript	**Lean theorem**	**Location**
Theorem 2.1	`w_half`	Optimal.lean:102
Theorem 3.1 (joint)	`above_half`	Above.lean:108
Corollary 3.2	`above_linear_rec`	AboveLimit.lean:71
Theorem 3.3 (existence)	`above_limit_exists`	AboveLimit.lean:89
Theorem 3.3 (lower bound)	`above_limit_ge`	AboveLimit.lean:133
Proposition 4.2	`deficit_succ`	Perturbation.lean:301
Proposition 4.3	`deficit_first_order`	Perturbation.lean:2503
Example 4.4 ($c_1 = 1$)	`c_one`	Perturbation.lean:76
Example 4.4 ($c_2 = \frac{3}{2}$)	`c_two`	Perturbation.lean:119
Example 4.4 ($c_3 = \frac{27}{16}$)	`c_three`	Perturbation.lean:169
Example 4.4 ($c_4 = \frac{111}{64}$)	`c_four`	Perturbation.lean:186
Example 4.4 ($c_5 = \frac{3555}{2048}$)	`c_five`	Perturbation.lean:211
Example 4.4 ($c_6 = \frac{113337}{65536}$)	`c_six`	Perturbation.lean:244
Lemma 4.5 (low)	`suffMin_collapse_low`	Perturbation.lean:1501
Lemma 4.5 (high)	`suffMin_collapse_high`	Perturbation.lean:1507
Lemma 4.6	`c_ge_27_16_full`	Perturbation.lean:1483
Lemma 4.7	`c_strict_anti_from_five`	Perturbation.lean:1488
Proposition 4.8	`c_linear_rec`	Perturbation.lean:1495
Identity (13)	`alg_id`	Perturbation.lean:932
$\sum_{n\geq 13} B_n < \frac{1}{8}$ (§4.3)	`B_tail_bound`	Perturbation.lean:864
$\varepsilon_{12} \geq \frac{1}{60}$ (§4.3)	`c_twelve_buffer`	Perturbation.lean:2995
$\sum_{n\geq 13} \delta_n \leq \frac{1}{200}$ (§4.3)	`delta_tail_bound`	Perturbation.lean:889
Cumulative ε-bound (§4.3)	`cum_eps_bound`	Perturbation.lean:949
Corollary 4.9	`c_limit_exists`	Perturbation.lean:1517
Theorem 4.10	`c_limit_formula`	Perturbation.lean:2065
Corollary 4.11 (i)	`w_gap_first_order`	Perturbation.lean:2811
Corollary 4.11 (ii)	`w_local_min_at_five`	Perturbation.lean:2847
Corollary 4.11 (iii)	`no_first_order_local_max`	Perturbation.lean:2948

Table 3: Correspondence between manuscript results and their Lean formalization. Links point to the exact line of commit `c60bcd3` on the repository's `main` branch.